

Affine extension from open balls

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_ball`

theorem

Applies the open connected theorem to two positive-radius balls.

Statement

Let E, F be real normed spaces, $c \in E$, $d \in F$, and $r, R > 0$. For a bijective isometry $f : B(c, r) \rightarrow B(d, R)$ there exists a unique surjective real affine isometry $A : E \rightarrow F$ satisfying $A(x) = f(x)$ for all $x \in B(c, r)$.

Assumptions

Both radii are positive. They are not assumed equal. No nonzero-dimension or completeness hypothesis is imposed on the spaces.

Conclusion

The specified isometry of open balls has exactly one ambient affine-isometry extension.

Proof route

Check openness and nonempty connectedness of the source ball, then substitute the two balls into the open connected theorem.

▼ Proof steps

1. The two balls are open. Since $r > 0$, $c \in B(c, r)$, and this ball is convex: for x, y in it and $0 \leq a \leq 1$, $\|(1-a)x + ay - c\| \leq (1-a)\|x - c\| + a\|y - c\| < r$. Thus it is nonempty and connected. Positivity of R also makes the target ball nonempty and connected, although target connectedness is not an independent input to the next theorem.
2. Apply Open connected extension theorem with $s = B(c, r)$, $t = B(d, R)$ and the given f . The openness and source connectedness just verified supply its hypotheses. Its single existence-and-uniqueness conclusion gives $A(x) = f(x)$ for every source-ball point, exactly as required.

Main citations

- Exact theorem and proof
- Open connected extension theorem

Lean source signature (exact)

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theorem existsUnique_affineExtension_ball
  {c : E} {d : F} {r R : ℝ} (f : ball c r ≈i ball d R)
  (hr : 0 < r) (hR : 0 < R) :
  ∃! A : E ≈a i[ℝ] F,
    ∀ x : ball c r, A (x : E) = ((f x : ball d R) : F)
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In the source	Mathematical meaning
{c : E} {d : F} {r R : ℝ}	The two centers and two real radii in real normed spaces E, F .
(f : ball c r ≈ _i ball d R)	The distance-preserving bijection $f : B(c, r) \rightarrow B(d, R)$ between open balls.
(hr : 0 < r) (hR : 0 < R)	The two strict inequalities $r > 0, R > 0$; no equality of radii is assumed.
∃! A : E ≈ ^{a i} [ℝ] F	A unique surjective real affine isometry of the ambient spaces satisfying the next clause.
∀ x : ball c r, A (x : E) = ((f x : ball d R) : F)	For every x with $\ x - c\ < r$, the same A has value $f(x)$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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